

EvoSensory

MINI MANUAL · QUICK START & REFERENCE

Multi-sensory rhythmic timing training, built by a pediatric OT. This guide covers everything you need to run your first session, understand the data, and document it.

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THE EVIDENCE

Why Rhythmic Timing Training Works

Rhythmic timing training has a substantial evidence base built across more than three decades of research in neuroscience, occupational therapy, physical therapy, and special education. The brain's internal timing system — involving the cerebellum, basal ganglia, supplementary motor area, and prefrontal cortex — can be trained through structured, repetitive practice synchronizing movement to an external beat.

When this system is less efficient, the effects are visible in daily life: impulsivity, bilateral coordination difficulty, slow processing speed, reading challenges, handwriting struggles, and poor motor sequencing. EvoSensory measures timing errors to the millisecond and makes them documentable in a way a therapist watching cannot achieve with the naked eye.

Area	Research Finding	EvoSensory Connection
ADHD & Attention	Children with ADHD show measurable timing deficits independent of inattention. Structured rhythmic training reduces impulsivity and improves sustained attention through cerebellar-prefrontal pathways.	Early-tap tendency + session trend (drifting earlier) map directly onto ADHD timing profiles.
DCD & Bilateral Coordination	Rhythmic timing training improves alternating limb timing and interlimb synchronization in DCD. Improvements transfer to functional bilateral tasks.	Late-tap tendency is the characteristic DCD pattern. Levels 1-2 directly train bilateral alternation.
Reading & Phonological Processing	Temporal processing underlies phoneme discrimination. Children with dyslexia show significant timing deficits. Rhythmic training improves reading fluency.	Cognitive module — vocabulary spelled by syllable on the beat — directly targets phonological timing.
Motor Planning & Praxis	Sequenced rhythmic training enhances motor program formation and execution speed in praxis deficits.	Levels 3-5 train increasingly complex sequences. Drift analysis tracks sequencing under sustained demand.
TBI & Acquired Injury	Post-TBI timing deficits correlate with processing speed and cognitive-motor outcomes. Rhythmic training shows measurable improvement.	Session trend drifting late may indicate cognitive fatigue during the session itself — useful diagnostic data.
Parkinson's Disease	Rhythmic Auditory Stimulation (RAS) is one of the most supported non-pharmacological PD interventions — documented improvements in gait cadence and motor timing.	Spatial audio + wired headphones provides auditory entrainment mechanism central to RAS. Motion module adds gait movement training.
Stroke Rehabilitation	Rhythmic cuing improves motor timing and bilateral retraining in post-stroke upper and lower extremity rehabilitation.	Motion module (clapping, marching) and unilateral modality settings support targeted stroke rehab protocols.
Handwriting	Graphomotor timing is one of the strongest predictors of handwriting legibility and speed. Timed rhythmic practice produces transfer effects to handwriting fluency.	Classic levels 1-3 build bilateral and timing foundations that transfer to handwriting. Deviation data directly reflects graphomotor timing quality.

Why millisecond precision matters

8ms early versus 62ms early is invisible to the human eye but clinically significant. The 8ms patient has an established internal clock. The 62ms patient has a systematic anticipatory pattern that mirrors impulsivity in daily functioning. EvoSensory makes that distinction visible and documentable in every session.

STEP BY STEP

Quick Start

1 Open the app

Go to **www.evolve-ot.com/evosensory.html** in Chrome. Bookmark it or tap Share → Add to Home Screen for easy access next session.

2 Create a patient profile

Tap **+ New Player**. Enter name or initials, select age category (Pediatric, Adult, or Older Adult), and choose training modality. Starting BPM is set automatically. On phones and tablets, most children tap with one hand — select the single-hand option if that applies.

3 Choose a module and level

Select **Classic → Level 1**. Adjust BPM with + / −. Use Classic with every new patient.

4 Confirm hardware

A setup screen appears before each session. Check what's connected — wired headphones, device vibration, XAC pillow. Only what you confirm is documented in the SOAP note.

5 Tap with the beat

Tap the large button, tap the canvas, or press **spacebar** on desktop. One tap per beat.

6 Generate SOAP note

When the session ends, tap **Generate SOAP Note**. Pre-populated draft ready to copy or print. Edit before pasting into your EMR.

Use wired headphones — most important upgrade

Spatial audio is the core clinical mechanism — early taps go to the left ear, late taps to the right. Bluetooth adds 100–200ms lag that defeats this. A \$10 wired earbud is all you need.

Patient Profiles

Category	Hands BPM	Feet BPM	Notes
Pediatric (0–17)	60 BPM	52 BPM	Standard starting point
Adult (18–64)	66 BPM	56 BPM	Slightly faster baseline
Older Adult (65+)	52 BPM	44 BPM	Wider hit window applied automatically

Feet start 8–10 BPM slower: hip → knee → ankle → floor is a longer motor sequence than a hand tap. On phones/tablets, verify bilateral hand use or select the appropriate single-hand modality.

THE FIVE MODULES

Choosing the Right Module

Module	What It Trains	Requires
 Classic	Core timing, bilateral coordination, spatial audio + haptic feedback, 6 progressive levels. Start every patient here.	Any device
 Oculomotor	Visual tracking synced to beat — pursuits, saccades, depth pulse, gaze hold. Tap-based motor proxy.	Any device
 Cognitive	Vocabulary and math spoken on the beat — phonological processing and number fluency.	Any device
 Motion	Camera detects clapping, knee taps, marching, and jumps timed to the beat.	Chrome on computer
 Active Games	Body movement controls 5 games — catch stars, pop balloons, kick soccer, duck/jump, balance zones.	Chrome on computer

Classic — Level Progression

Level	Pattern	Advance When
1	Alternating L/R	85%+ accuracy, 2 sessions
2	Alternating with rests	85%+ accuracy, 2 sessions
3	Longer held beats	80%+ accuracy, stable trend
4	Syncopated patterns	80%+ accuracy, <50ms deviation
5	Irregular + rests	80%+ accuracy, consistent timing
6	Full complexity	Clinical ceiling — maintain for consolidation

Motion — Sub-Modes (Chrome on computer required)

Clapping (bilateral UE, waist-up) · Knee Taps/Midline Crossing (waist-head) · Marching (full body) · Jumps (full body + clearance)

Active Games — Sub-Modes (Chrome on computer required)

Catch Stars (lateral stepping, vestibular) · Pop Balloons (overhead reach) · Kick Soccer (unilateral foot) · Duck & Jump (motor planning) · Side Steps & Balance (static/dynamic balance)

BPM progression rule

Increase tempo by 4–8 BPM at a time. Accuracy at a slow, steady tempo is the foundation for everything else. If accuracy falls below 50%, reduce BPM before changing the level.

READING THE DATA

Understanding Your Results

1. Accuracy — Were they hitting the beat?

% of beats hit within the acceptable time window. Reflects engagement, attention, and task level. High accuracy does not mean good timing.

2. Timing Deviation — Where exactly did taps land?

ms (milliseconds) from the beat. Lower deviations are better. Zero = perfectly on the beat.

(+) = early (before the beat) (–) = late (after the beat)

Accuracy and deviation are not the same thing

A patient can score 95% accuracy but have +55ms average deviation — hitting the window every time, but always early. That is a separate, clinically meaningful finding that accuracy alone does not capture.

3. Session Trend — Did timing change during the session?

Tracks whether errors grew, shrank, or held steady as the session progressed. Drifting earlier may indicate difficulty sustaining inhibitory control. Drifting later may indicate attention fatigue.

Clinical Interpretation Scale

ACCURACY

Range	Interpretation	Action
85–100%	Functional range — skill established	Advance level or increase BPM by 4–8
70–84%	Emerging — moderate consistency	Continue; target 85% across 2 sessions
50–69%	Below threshold — significant difficulty	Adjust supports; consider slower BPM
<50%	Well below threshold	Reduce BPM 8–10 or drop one level

TIMING DEVIATION (ABSOLUTE VALUE — LOWER IS BETTER)

Range	Interpretation	Clinical Relevance
0–20ms	Excellent — normal limits	Strong internal clock
21–40ms	Good — mild variability	Typical in active development
41–70ms	Moderate difficulty	Likely affecting handwriting, bilateral tasks
71–100ms	Significant difficulty	Document functional impact; intervention warranted
100ms+	Severe difficulty	Reduce demands; prioritize sensory feedback supports

DIRECTION & SESSION TREND

Pattern	May Indicate
Consistently + (early)	Impulsivity, response inhibition difficulty, anticipatory motor pattern (ADHD, anxiety)
Consistently – (late)	Processing speed delay, motor initiation difficulty, low arousal (DCD, TBI, fatigue)
Stable across session	Consistent regulation — patient sustained the task without deterioration
Drifting early over time	Mounting impulsivity or difficulty sustaining inhibitory control
Drifting late over time	Attention fatigue or processing slowdown under sustained demand

Key documentation point

Always report both accuracy and deviation. Connect the deviation direction to a functional impact in the patient's daily occupations — handwriting, turn-taking, dressing, gait, reading rhythm.

DOCUMENTATION

SOAP Note Guide

Tap **Generate SOAP Note** on the results screen. Pre-populated with all session data. Includes a "how to read this note" guide at the bottom for reviewers unfamiliar with the tool.

Section	Content
S	Blank — therapist fills in observations, engagement, patient report
O	Accuracy %, deviation with direction, session trend, level, BPM, modality, hardware confirmed, mobile caveat if applicable
A	Plain-language interpretation with functional connection prompts and progression status
P	Specific next-session BPM and level recommendation based on this session's data

Always edit before use

Draft only. The A section has fill-in prompts — review and personalize before pasting into your EMR or submitting for billing.

Settings Reference

Beat Sound

Single Beat · Double Beat · 1-2 Spoken (says "one" and "two" — good for young children)

Blind Training Mode

Fades visuals after 10s — trains audio/haptic timing only, no visual anchor

Therapist View

Shows ms deviation and clinical notes in real time during session

Session Setup Prompt

Hardware confirmation before each session — turn off once workflow is established

Device Haptics / XAC

Device vibration + XAC pillow bilateral rumble (left = early, right = late)

Metronome Toggle

Turn off to test internal timing — patient taps without auditory cue

Hardware & Calibration

Setup	Cost	Adds
Any device + browser	\$0	Full training, SOAP notes, multi-user profiles
+ Wired earbuds	~\$10	Spatial audio: left ear = early, right ear = late
+ XAC + haptic cushion	~\$150-200	Directional whole-body haptic feedback
Chrome on computer	—	Motion, Active Games, eye tracking

Eye Tracking Calibration (Chrome on Windows/Mac/Android only)

Enable toggle in Oculomotor panel → tap Play → 9-point calibration grid appears → look at each dot and tap it → tap Start. Gaze crosshair on canvas. Accuracy ± 60 -80px (screening-level). SOAP note includes disclosure automatically.

Camera Positioning (Motion & Active Games)

Positioning guide appears in the setup overlay for each sub-mode. Start button enables automatically when required landmarks are detected — full-body modes wait for both ankles visible.